Project 4: Intelligent Document Processing (Extraction & Automation)

Background

Finance and operations teams spend hours manually keying data from invoices, purchase orders, receipts, and forms into their systems. The work is slow, costly, and error-prone — and it scales badly as volume grows.

Xquisite AI wants students to build an intelligent document-processing pipeline that uses a multimodal LLM to read a document and output clean, structured data, with validation and a human review step before it is exported.

Scope & Deliverables

- Upload of a document (PDF or image) and LLM extraction of structured fields per document type (invoice, PO, receipt)
- Validation rules (totals reconcile, required fields present, format checks) that flag issues
- A human-in-the-loop review screen to correct and approve extracted data
- Export to CSV/JSON and a mock API endpoint
- A field-level accuracy evaluation against ground-truth labels
- A cost-benefit business case versus manual data entry

Skills Students Will Build

Skill area	What students practice
Prompt Engineering	Instructing the model to return exactly the fields needed
Multimodal LLMs	Reading scanned documents and images (vision)
Structured Outputs	Schema-constrained extraction (pydantic / JSON schema, function calling)
Validation Logic	Business-rule checks and confidence flagging
Human-in-the-Loop Design	Review, correct, approve workflow
Entrepreneurship	Cost-benefit and automation ROI

Work Steps

1. **Define Extraction Schemas —** Model each document type as a structured schema (pydantic), e.g. vendor, date, line items, totals.
2. **Document Loading —** Convert PDFs/images to a model-readable form (PyMuPDF / pdf2image).
3. **Extraction —** Prompt a multimodal LLM to return structured output; use a vision model for scans.
4. **Validation & Flagging —** Apply business rules (e.g. line items must sum to the total) and flag low-confidence fields.
5. **Review UI —** Show the document next to the extracted fields for edit-and-approve.
6. **Export & API —** Export approved data to CSV/JSON and a mock API endpoint.
7. **Evaluation —** Measure field-level precision/recall against the labelled ground truth.

Dummy Data Provided

≈200 synthetic documents (invoices, purchase orders, and receipts) as PDF/PNG, plus ground_truth.csv with the correct labelled fields for evaluation.

doc_id	doc_type	vendor	total_amount
DOC-001	invoice	PT Sumber Makmur	12,450,000
DOC-002	purchase_order	CV Mitra Teknik	3,900,000
DOC-003	receipt	Toko Sentosa	275,000

Additional columns: invoice_date, due_date, line_items (JSON), tax_amount, currency.

Recommended Tools & Technologies

- A multimodal LLM: OpenAI GPT-4o (vision) or a vision-capable local model via Ollama
- PyMuPDF or pdf2image for document loading
- pydantic for schema-constrained extraction
- Streamlit for the review UI
- JSON schema / function calling for structured output